Initial nurse workforce calculation

Number of nurses in Alberta: 38,900

Percentage of nurses working in hospitals: 67%

Number of nurses working in hospitals in Alberta: $38,900 \times 0.67 = 26,063$

Percentage of nurses working in COVID-unit: 26%

Initial number of nurses working in COVID-unit in Alberta: $26,063 \times 0.26 = 6776$

Initial number of nurses working in other-unit in Alberta: $26,063 - 6776 = 19,287$

Table 1: Weekly community infection risk calculation

	Alpha	**Omicron**
Community seroprevalence from data	August 2020: 0.00749 December 2020: 0.01996 Difference: 0.01247	December 2021: 0.1294 February 2022: 0.2489 Difference: 0.1195
Weekly increase in community seroprevalence	$\int_{20}^{40} mx\,dx = 0.01247$ $m = 2.0783 \times 10^{-5}$	$\int_{91}^{97} mx\,dx = 0.1195$ $m = 2.1188 \times 10^{-4}$
Weekly community infection risk	$\theta_{20} = \int_{20}^{21} mx\,dx = 0.0004260515$ $\theta_{21} = \int_{21}^{22} mx\,dx = 0.0004468345$ $\theta_{22} = \int_{22}^{23} mx\,dx = 0.0004676175$ $\theta_{23} = \int_{23}^{24} mx\,dx = 0.0004884005$ $\theta_{24} = \int_{24}^{25} mx\,dx = 0.0005091835$ $\theta_{25} = \int_{25}^{26} mx\,dx = 0.0005299665$ $\theta_{26} = \int_{26}^{27} mx\,dx = 0.0005507495$ $\theta_{27} = \int_{27}^{28} mx\,dx = 0.0005715325$ $\theta_{28} = \int_{28}^{29} mx\,dx = 0.0005923155$ $\theta_{29} = \int_{29}^{30} mx\,dx = 0.0006130985$ $\theta_{30} = \int_{30}^{31} mx\,dx = 0.0006338815$ $\theta_{31} = \int_{31}^{32} mx\,dx = 0.0006546645$	$\theta_{91} = \int_{91}^{92} mx\,dx = 0.01938822$ $\theta_{92} = \int_{92}^{93} mx\,dx = 0.01959590$ $\theta_{93} = \int_{93}^{94} mx\,dx = 0.01980338$ $\theta_{94} = \int_{94}^{95} mx\,dx = 0.02001006$ $\theta_{95} = \int_{95}^{96} mx\,dx = 0.02021674$ $\theta_{96} = \int_{96}^{97} mx\,dx = 0.02042342$

	$\theta_{32} = \int_{32}^{33} mx\,dx = 0.0006754475$ $\theta_{33} = \int_{33}^{34} mx\,dx = 0.0006962305$ $\theta_{34} = \int_{34}^{35} mx\,dx = 0.0007170135$ $\theta_{35} = \int_{35}^{36} mx\,dx = 0.0007377965$ $\theta_{36} = \int_{36}^{37} mx\,dx = 0.0007585795$ $\theta_{37} = \int_{37}^{38} mx\,dx = 0.0007793625$ $\theta_{38} = \int_{38}^{39} mx\,dx = 0.0008001455$ $\theta_{39} = \int_{39}^{40} mx\,dx = 0.0008209285$	

Table 2: Weekly Costs for Alpha (Aug - Dec 2020)

Week	Total Infected Nurses	Total Cost for 0% Redeployment	Total Cost for 50% Redeployment	Total Cost for 100% Redeployment
20-21	2.8869	8206.879	6839.066	5471.253
21-22	4.7925	13624.12	11353.43	9082.746
22-23	6.0957	17328.86	14440.71	11552.57
23-24	7.0312	19988.3	16656.91	13325.53
24-25	7.7419	22008.67	18340.56	14672.45
25-26	8.3131	23632.48	19693.73	15754.99
26-27	8.8007	25018.63	20848.86	16679.09
27-28	9.2426	26274.86	21895.72	17516.58
28-29	9.6556	27448.94	22874.12	18299.29
29-30	10.0574	28591.18	23825.98	19060.78
30-31	10.4596	29734.55	24778.79	19823.03
31-32	10.8546	30857.46	25714.55	20571.64
32-33	11.2523	31988.04	26656.7	21325.36

33-34	11.6613	33150.74	27625.62	22100.5
34-35	12.0990	34395.04	28662.53	22930.02
35-36	12.5667	35724.61	29770.51	23816.41
36-37	13.0801	37184.11	30986.76	24789.41
37-38	13.6632	38841.74	32368.12	25894.5
38-39	14.3019	40657.44	33881.2	27104.96
39-40	14.9551	42514.36	35428.63	28342.91
Total Cost		567171	472642.5	378114

Table 3: Weekly Costs for Omicron (Dec 2021 – Feb 2022)

Week	Total Infected Nurses	Total Cost for 0% Redeployment	Total Cost for 50% Redeployment	Total Cost for 100% Redeployment
91-92	131.3746	373471.7	311226.4	248981.1
92-93	214.2678	609120.5	507600.4	406080.3
93-94	267.4039	760175.8	633479.8	506783.9
94-95	303.8044	863655.1	719712.6	575770.1
95-96	332.3866	944908.6	787423.9	629939.1
96-97	357.2265	1015523	846269.6	677015.7
Total Cost		4566855	3805713	3044570

Table 4: Weekly Infected and Backfill Nurses (0% Redeployment, Alpha):

Week	Total Infected Nurses	Total Overtime Hours	Nurses Required for 4 hours Overtime	Nurses Required for 8 hours Overtime	Nurses Required for 20 hours Overtime
20-21	2.89	115	29	14	6
21-22	5.91	192	48	24	10
22-23	6.2	244	61	30	12
23-24	6.48	281	70	35	14
24-25	6.76	310	77	39	15
25-26	7.04	333	83	42	17

26-27	7.32	352	88	44	18
27-28	7.6	370	92	46	18
28-29	7.89	386	97	48	19
29-30	8.17	402	101	50	20
30-31	8.45	418	105	52	21
31-32	8.73	434	109	54	22
32-33	9.01	450	113	56	23
33-34	9.29	466	117	58	23
34-35	9.58	484	121	60	24
35-36	9.86	503	126	63	25
36-37	10.14	523	131	65	26
37-38	10.42	547	137	68	27
38-39	10.7	572	143	72	29
39-40	10.98	598	150	75	30

Table 5: Weekly Infected and Backfill Nurses (0% Redeployment, Omicron):

Week	Total Infected Nurses	Total Overtime Hours	Nurses Required for 4 hours Overtime	Nurses Required for 8 hours Overtime	Nurses Required for 20 hours Overtime
91-92	131.37	5255	1314	657	263
92-93	264.17	8571	2143	1071	429
93-94	267.04	10696	2674	1337	535
94-95	269.91	12152	3038	1519	608
95-96	272.78	13295	3324	1662	665
96-97	275.65	14289	3572	1786	714

Table 6: Weekly Infected and Backfill Nurses (50% Redeployment, Alpha):

Week	Total Infected Nurses	Total Overtime Hours	Nurses Required for 4 hours Overtime	Nurses Required for 8 hours Overtime	Nurses Required for 20 hours Overtime
20-21	2.89	58	14	7	3
21-22	5.91	96	24	12	5
22-23	6.2	122	30	15	6
23-24	6.48	141	35	18	7
24-25	6.76	155	39	19	8
25-26	7.04	166	42	21	8

26-27	7.32	176	44	22	9
27-28	7.6	185	46	23	9
28-29	7.89	193	48	24	10
29-30	8.17	201	50	25	10
30-31	8.45	209	52	26	10
31-32	8.73	217	54	27	11
32-33	9.01	225	56	28	11
33-34	9.29	233	58	29	12
34-35	9.58	242	60	30	12
35-36	9.86	251	63	31	13
36-37	10.14	262	65	33	13
37-38	10.42	273	68	34	14
38-39	10.7	286	72	36	14
39-40	10.98	299	75	37	15

Table 7: Weekly Infected and Backfill Nurses (50% Redeployment, Omicron):

Week	Total Infected Nurses	Total Overtime Hours	Nurses Required for 4 hours Overtime	Nurses Required for 8 hours Overtime	Nurses Required for 20 hours Overtime
91-92	131.37	2627	657	328	131
92-93	264.17	4285	1071	536	214
93-94	267.04	5348	1337	669	267
94-95	269.91	6076	1519	760	304
95-96	272.78	6648	1662	831	332
96-97	275.65	7145	1786	893	357